

NEW RIVER VALLEY, VA, USA

WMO: 724116

Lat: 37.133N

Lon: 80.683W

Elev: 642

StdP: 93.85

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13868

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.0	-18.8	0.8	-6.8	-16.3	1.0	-5.6	12.0	0.8	11.0	0.9	3.5	300	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.4	21.5	31.2	21.4	29.9	21.1	23.8	29.1	23.1	27.9	22.5	27.2	2.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.5	25.9	21.9	18.0	25.5	21.2	17.1	24.8	74.5	29.3	72.1	27.9	69.6	27.2	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.3	7.3	-14.3	34.2	3.2	1.6	-16.7	35.3	-18.5	36.2	-20.3	37.1	-22.7	38.3		
(5)			-15.1	24.8	3.1	0.8	-17.3	25.3	-19.1	25.8	-20.9	26.3	-23.2	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	1.9	3.6	7.6	13.0	17.7	21.8	23.6	22.9	19.4	13.6	7.7	3.7		
	DBStd	8.81	5.71	5.46	5.39	4.66	4.00	2.88	2.37	2.37	3.41	4.32	4.80	4.95		
	HDD10.0	903	258	189	111	23	2	0	0	0	17	100	204			
	HDD18.3	2448	510	413	332	167	61	7	1	28	154	319	454			
	CDD10.0	2031	6	9	37	113	241	354	420	399	283	129	31	8		
	CDD18.3	535	0	0	1	7	42	111	163	142	61	8	1	0		
	CDH23.3	4497	0	1	10	110	394	924	1350	1128	489	86	4	0		
	CDH26.7	1434	0	0	1	18	89	296	491	386	139	14	0	0		
(14) Wind	WSAvg	2.5	3.4	3.3	3.3	3.2	2.4	1.9	1.6	1.4	1.6	2.1	2.6	2.8		
(15) Precipitation	PrecAvg	1004	81	63	95	80	109	100	107	84	78	63	66	78		
	PrecMax	1358	170	161	183	129	228	217	181	155	226	170	118	159		
	PrecMin	775	33	20	27	23	49	27	44	37	7	2	11	32		
	PrecStd	141	36	31	37	29	51	48	36	31	48	44	31	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	21.1	24.9	28.8	31.2	33.0	34.0	33.1	32.3	28.2	23.6	19.1		
		MCWB	13.9	14.2	14.9	16.5	19.7	21.6	21.8	21.4	19.2	18.6	15.2	14.4		
	2%	DB	16.1	17.6	22.2	26.3	28.9	31.4	32.4	32.2	30.0	26.0	20.9	16.4		
		MCWB	12.4	11.6	13.6	16.1	19.5	21.6	22.0	21.7	20.1	17.4	13.5	13.1		
	5%	DB	13.0	14.9	18.9	24.0	27.5	29.9	31.1	30.6	28.0	23.3	17.8	14.0		
		MCWB	9.9	10.6	12.0	15.1	18.7	21.1	21.8	21.7	19.7	16.3	13.0	11.3		
	10%	DB	11.0	12.4	17.0	22.2	25.9	28.0	29.2	28.8	26.3	21.3	16.1	12.1		
		MCWB	8.1	8.7	11.2	14.3	18.4	20.3	21.4	21.1	19.1	15.7	11.7	9.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	15.9	16.8	19.1	22.0	24.2	24.7	24.3	23.2	21.6	17.4	16.4		
		MCDB	17.5	19.3	22.1	25.5	27.4	30.6	30.5	29.9	27.4	25.0	19.9	18.3		
	2%	WB	13.4	13.7	15.1	17.8	21.0	23.0	24.0	23.5	22.1	19.7	15.7	14.1		
		MCDB	15.6	16.3	19.4	23.7	26.2	28.1	29.2	28.7	26.3	23.0	18.5	15.5		
	5%	WB	10.9	11.5	13.8	16.7	20.1	22.2	23.3	22.8	21.2	18.0	14.3	12.2		
		MCDB	12.8	13.9	17.8	21.9	25.2	27.4	28.2	27.6	25.4	21.3	17.2	13.7		
	10%	WB	8.0	9.0	12.2	15.3	19.3	21.4	22.6	22.1	20.4	16.7	12.5	9.7		
		MCDB	9.7	11.9	15.6	19.8	24.0	26.4	27.3	26.8	24.3	20.2	15.1	11.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.2	11.4	12.5	12.2	11.6	11.1	11.3	11.6	12.0	11.4	9.5		
		MCDBR	12.8	14.5	15.4	16.0	14.8	13.8	13.4	13.7	14.3	15.2	14.9	13.0		
	5% WB	MCWBR	9.2	9.7	8.6	7.9	7.1	5.5	5.1	5.3	5.9	7.7	9.4	9.5		
		MCWBR	11.2	12.2	12.6	13.4	12.5	12.0	11.1	11.4	11.3	11.7	11.6	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.307	0.332	0.373	0.414	0.441	0.473	0.479	0.377	0.344	0.309	0.293		
	taud		2.502	2.459	2.435	2.341	2.231	2.187	2.129	2.087	2.373	2.459	2.522	2.538		
	Ebn at Noon		901	931	934	908	871	844	813	798	873	875	877	881		
	Edh at Noon		80	94	106	123	140	146	154	157	112	93	78	72		
(44) All-Sky Solar Radiation	RadAvg		2.22	2.91	3.97	5.05	5.57	6.07	5.68	5.22	4.40	3.47	2.59	1.93		
	RadStd		0.19	0.35	0.26	0.41	0.46	0.43	0.35	0.27	0.43	0.43	0.25	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+82

Nomenclature: See separate page